BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 083 · Lesson 083 · Dijkstra: Shortest Paths with Non-negative Weights

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does relaxation do?

Q2. What is a weighted edge?

Q3. What condition does standard Dijkstra require?

Q4. Undirected graph edges A–B, A–C, B–D. Give BFS order from A visiting B before C.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Dijkstra — Non-negative weighted shortest paths.

tentative distance — Current best-known route cost.

relaxation — Replace distance if improved.

predecessor — Previous vertex on best path.

New learning

Observe — Dijkstra maintains tentative shortest distances from a start vertex. All edge weights must be non-negative.

Apply — Repeatedly finalise the unsettled node with smallest tentative distance, then relax outgoing edges to improve neighbour estimates.

Evaluate — Track predecessor nodes alongside distances to reconstruct the shortest route, not just its total cost.

Worked example

Find the shortest P→R route. Directed edges: P→Q = 3, P→R = 8, Q→R = 2.

1. Update neighbours — Start with dist(P)=0 . Settle P (smallest unsettled value). Relax P→Q: Q=3 , predecessor Q←P. Relax P→R: R=8 , predecessor R←P. Remaining unsettled distances: Q=3, R=8.

2. Improve the tentative route — Q=3 is the smallest unsettled distance. Via Q→R, the candidate is 3+2=5 . This beats R=8, so update R=5 and predecessor R←Q. Settled now: P,Q.

3. Reconstruct the actual path — R=5 is next, and is the target. Follow predecessor pointers R←Q←P ; reverse to obtain P→Q→R , total cost 5 . No vertex with a larger tentative distance was prematurely finalised.

We Do · Practise with guidance

Find shortest X→Z. Directed edges: X→Y = 4, X→Z = 10, Y→Z = 3.

```
Initial: X=0, Y=∞, Z=∞
Complete:
Settled vertex | dist(Y) | dist(Z) | predecessor of Z
X              |     ?   |     ?   |    ?
Y              |     ?   |     ?   |    ?
Z              |     ?   |     ?   |    ?
```

Settle X; calculate both tentative distances

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Start at X=0. Which neighbour gets distance 4? Which gets 10? Which unsettled vertex should Dijkstra select next?

Settle Y and relax Y→Z

After X: Y=4 (predecessor X), Z=10 (predecessor X). Settle Y=4; candidate cost to Z is 4+3=7 , so improve Z to 7 with predecessor Y.

Finish with the path, not just the cost

Settle Z=7. Predecessors: Z←Y←X . Reverse to X→Y→Z , total cost 7. The direct X→Z cost 10 is longer.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

From A in stated graph, give tentative distances after settling A and B. [6]

Question 2

Continue from C: give final shortest cost and predecessor chain for D. [6]

Question 3

Explain why negative edge weights can invalidate Dijkstra's finalisation rule, using a counterexample. [6]

Exit Ticket · Answer before checking

Dijkstra is guaranteed correct on graphs with arbitrary negative edge costs.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Updates a tentative distance via a cheaper path.

Q2. Connection with a cost.

Q3. Non-negative edge weights.

Q4. A,B,C,D.

You Do pages · Independent questions

Q1. After A B2,C5,D∞ [3]; after B C3,D6 [3].

Q2. C dist3 [1], relax C→D cost1 gives D4 [2], predecessors D←C←B←A [2], path cost4 [1].

Q3. Finalise an apparently shortest vertex [2], later negative edge can improve it [2], so settled assumption false [1], use suitable negative-weight method instead [1].

Exit Ticket · Model answer

False. The standard algorithm assumes non-negative weights.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: